HIGH PRESSURE FUEL PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

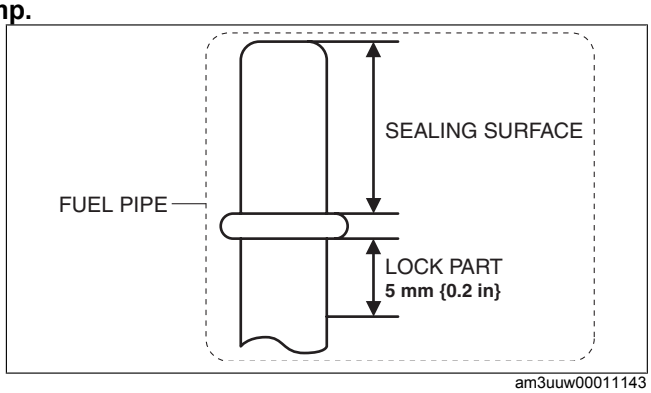
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Warning

- **Fuel is very flammable liquid. If fuel spills or leaks from the pressurized fuel system, it will cause serious injury or death and facility breakage. Fuel can also irritate skin and eyes. To prevent this, always complete the "Fuel Line Safety Procedure", while referring to "BEFORE SERVICE PRECAUTION".**
- **A person charged with static electricity could cause a fire or explosion, resulting in death or serious injury. Before draining fuel, make sure to discharge static electricity by touching the vehicle body.**

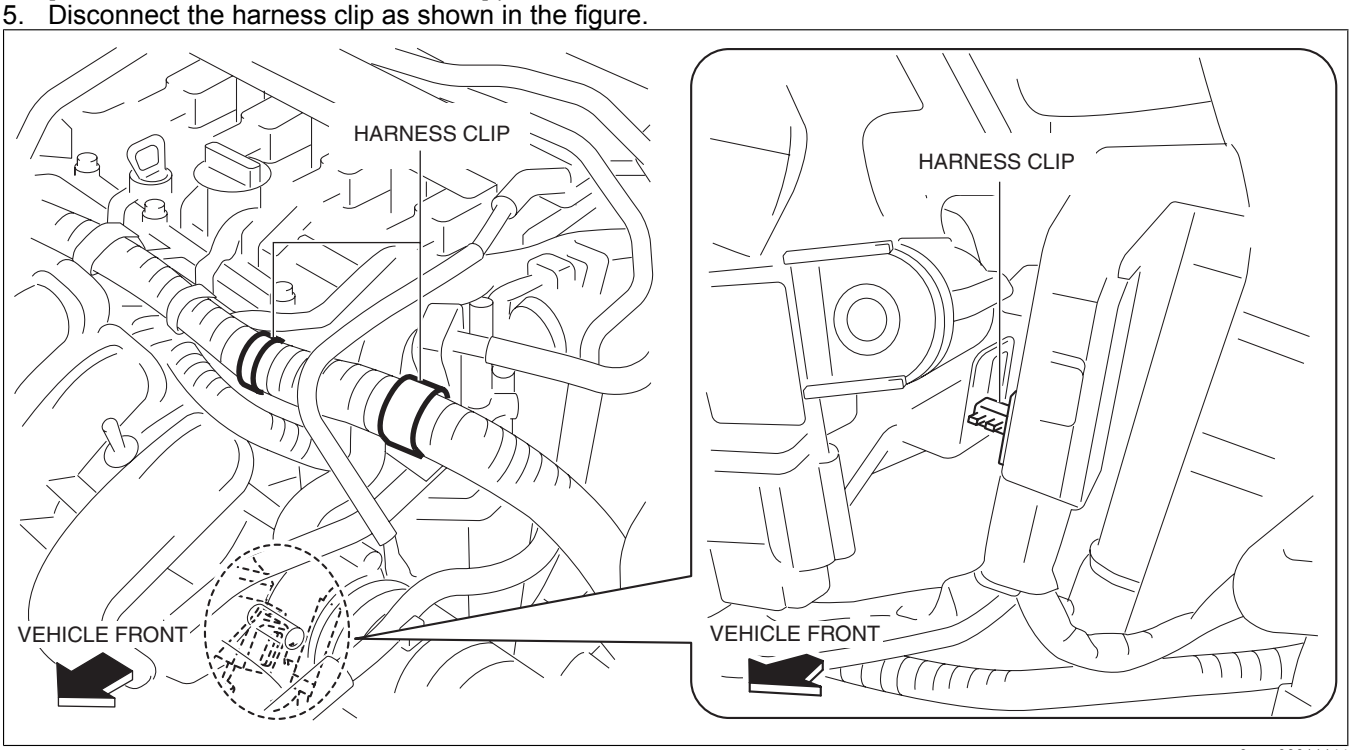
Caution

- **Do not disassemble the high pressure fuel pump.**
- **Do not damage the sealing surface or the lock part of the fuel pipe.**
- **Damage or deformation to the end of the high pressure line pipe may cause fuel leakage.**
- **Disconnecting/connecting the quick release connector without cleaning it may cause damage to the fuel pipe and quick release connector. Always clean the quick release connector joint area before disconnecting/connecting using a cloth or soft brush, and make sure that it is free of foreign material.**



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1. Complete the "BEFORE SERVICE PRECAUTION". (See BEFORE SERVICE PRECAUTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
3. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the air cleaner and air hose as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Disconnect the harness clip as shown in the figure.

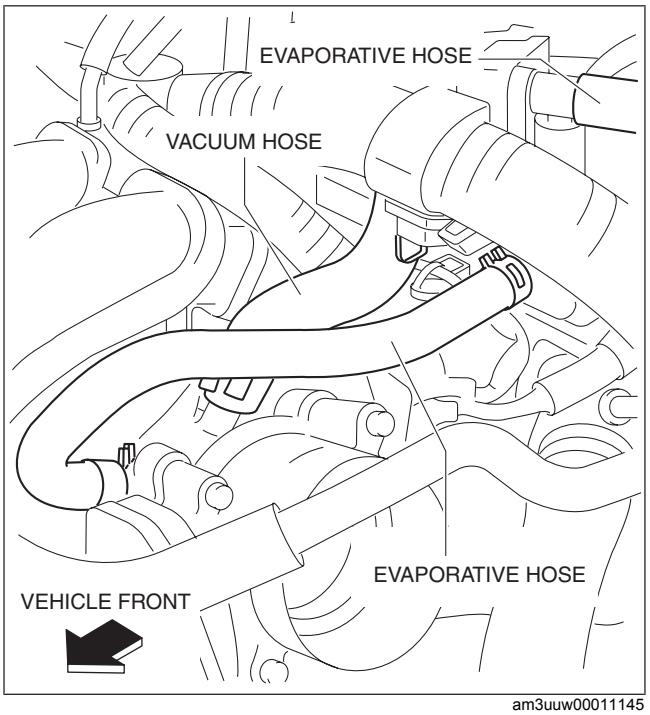


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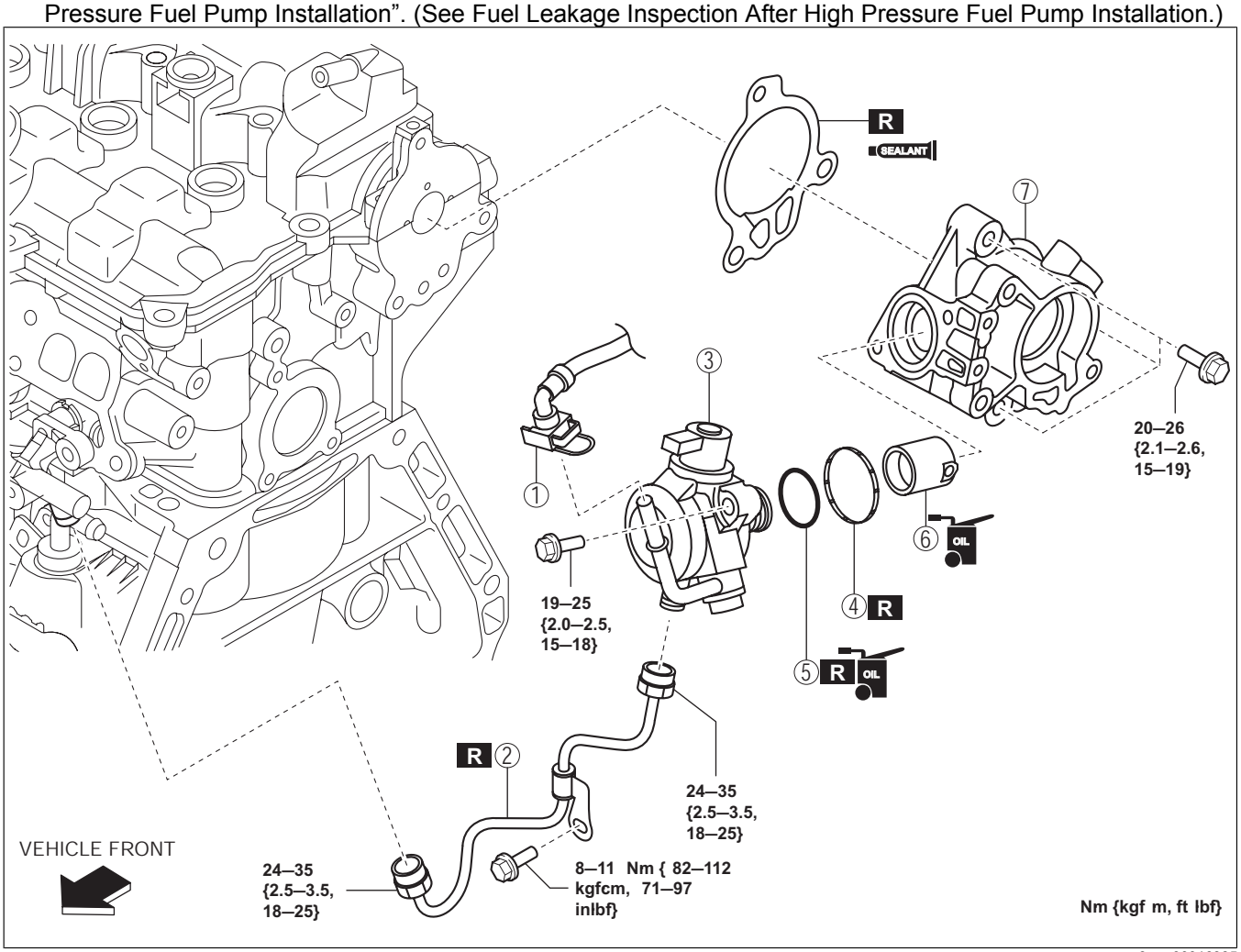
6. Disconnect the hoses as shown in the figure.
7. Disconnect the following parts connectors.
   - Purge solenoid valve
   - High pressure fuel pump
   - CMP sensor (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))
   - Intake CMP sensor (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
8. Remove the purge solenoid valve and catch tank and purge solenoid valve bracket as a single unit. (See PURGE SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9. Remove in the order indicated in the table.

Note
- Only when the high pressure fuel pump is removed, the rear housing does not have to be removed.

10. Install in the reverse order of removal.
11. Complete the “AFTER SERVICE PRECAUTION”. (See AFTER SERVICE PRECAUTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
12. Complete the “Fuel Leakage Inspection After High Pressure Fuel Pump Installation”. (See Fuel Leakage Inspection After High Pressure Fuel Pump Installation.)



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1	Quick release connector (See QUICK RELEASE CONNECTOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

2	High pressure fuel pipe (See High Pressure Fuel Pipe Removal Note.) (See High Pressure Fuel Pipe Installation Note.)

3	High pressure fuel pump (See High Pressure Fuel Pump Installation Note.)
4	Seal rubber
5	O-ring
6	Tension collar

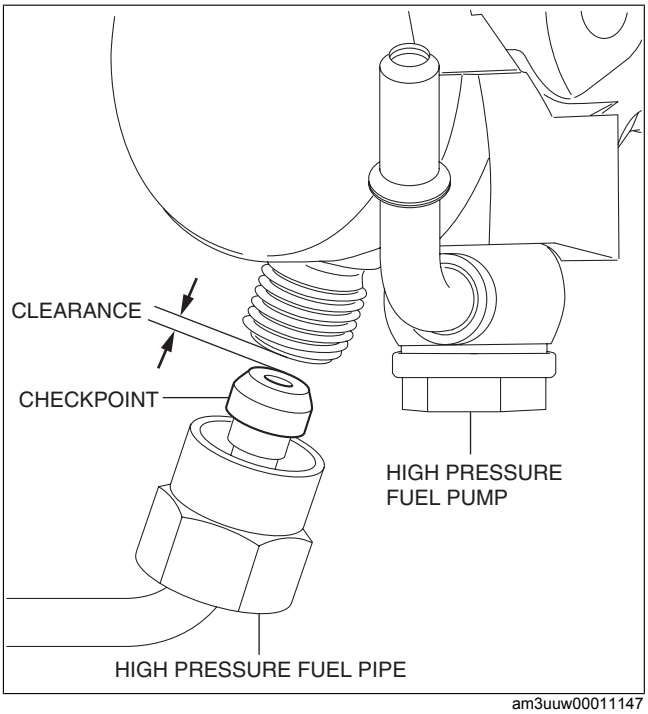
7	Rear housing (See Rear Housing Removal Note.) (See Rear Housing Installation Note.)

High Pressure Fuel Pipe Removal Note

Caution

- **When removing the high pressure fuel pipe, the end of the high pressure fuel pipe may be damaged because the clearance between the high pressure fuel pump and the high pressure fuel pipe is small. Be very careful when handling the end of the high pressure fuel pipe so as not to damage it.**
- **If the high pressure fuel pump is removed with the high pressure fuel pipe installed, the high pressure fuel pipe and fuel distributor may be damaged because of the excessive force caused by the spring force of the high pressure fuel pump. The high pressure fuel pipe and fuel distributor must be replaced if they are damaged. Always remove the high pressure fuel pipe before removing the high pressure fuel pump.**

1. Remove the high pressure fuel pipe.



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Rear Housing Removal Note

1. Remove the vacuum pump. (See VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Remove the rear housing.

Rear Housing Installation Note

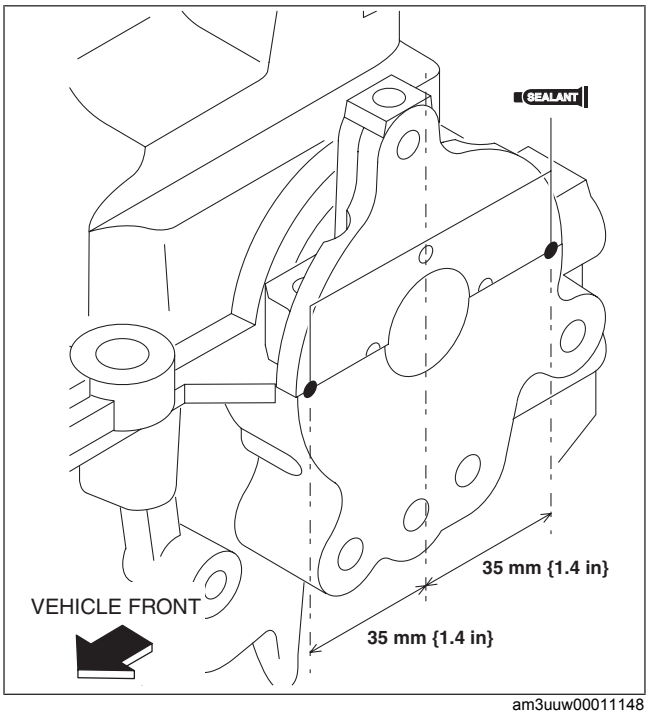
Note

- After applying silicone sealant, install the rear housing within 10 min. and then tighten all bolts within 5 min.

1. Apply silicone sealant to the areas shown in the figure.

   Thickness
   3.0—7.0 mm {0.12—0.27 in}

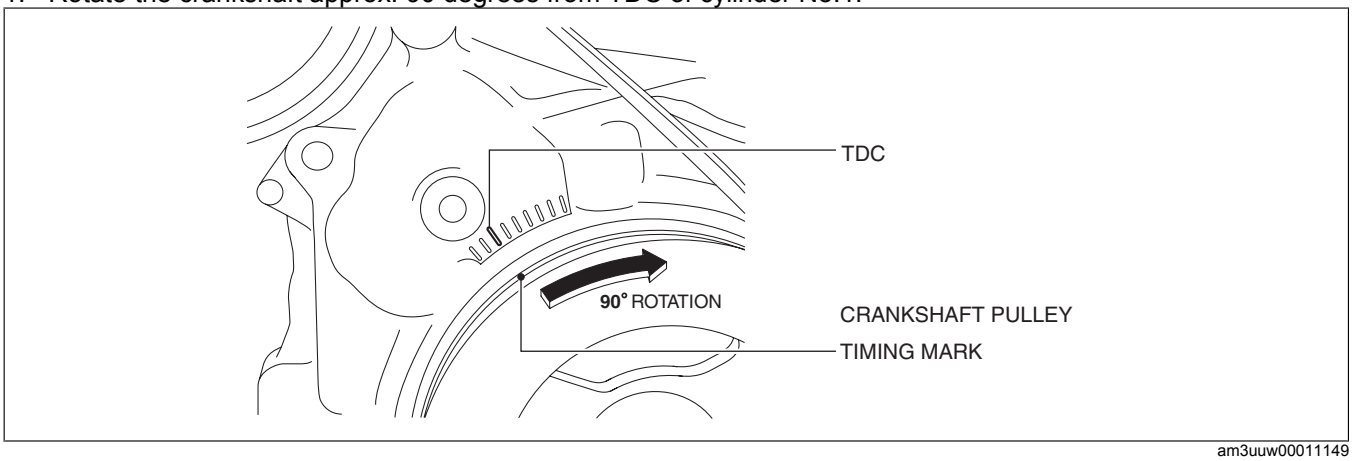
2. Install the rear housing.



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High Pressure Fuel Pump Installation Note

1. Rotate the crankshaft approx. 90 degrees from TDC of cylinder No.1.



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2. Insert the high pressure fuel pump into the rear housing.
3. Press in the high pressure fuel pump until the O-ring cannot be seen.
4. Temporarily tighten the high pressure fuel pump installation bolt on one side until its head contacts the high pressure fuel pump.
5. Tighten the high pressure fuel pump installation bolt which is not tightened in Step 4.

   High pressure fuel pump installation bolt tightening torque
   19—25 N·m {2.0—2.5 kgf·m, 15—18 ft·lbf}

6. Tighten the high pressure fuel pump installation bolt which was tightened in Step 4.

   High pressure fuel pump installation bolt tightening torque
   19—25 N·m {2.0—2.5 kgf·m, 15—18 ft·lbf}

High Pressure Fuel Pipe Installation Note

Caution

- **Damage or deformation to the end of the high pressure fuel pipe may cause fuel leakage. When installing the high pressure fuel pipe to the high pressure fuel pump, be very careful while handling the end of the high pressure fuel pipe.**
- **If parts are exposed to fuel, their function may be decreased or a fire may occur. Always wipe off any leaked fuel using a clean rag.**

- **Do not let foreign materials enter into the high pressure fuel pipe.**
- **Install a new high pressure fuel pipe because it cannot be reused.**

1. Temporarily tighten the nuts on both ends.
2. Temporarily tighten the high pressure fuel pipe bracket installation bolt.
3. Tighten the nuts on both ends.

   High pressure fuel pipe tightening torque
   24—35 N·m {2.5—3.5 kgf·m, 18—25 ft·lbf}

4. Tighten the high pressure fuel pipe bracket installation bolt.

   High pressure fuel pipe bracket installation bolt tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

Fuel Leakage Inspection After High Pressure Fuel Pump Installation

1. Verify that the high pressure fuel pump is assembled securely.
2. Verify that there is no fuel leakage around the high pressure fuel pump.